# Rudra D. Patel

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## EDUCATION

**Virginia Polytechnic Institute and State University**

M.Sc. (Thesis) Aerospace Engineering — **Overall GPA: *3.95***

B.Sc. Aerospace Engineering (Minor – Computer Science) — **Major GPA: *3.33***

## SKILLS

Software: SIMULINK, Stateflow, MATLAB, Mathematica, StarCCM+, ANSYS, React Js, LabView, Eclipse
Programming: Java, C, C++, HTML, CSS, JavaScript, Linux, Python
3-D Modelling: Autodesk Inventor, SolidWorks, CATIA, Siemens NX
Miscellaneous: Github, DOORS (DXL), JIRA, Teamcenter

## EXPREIENCE

Honda Aircraft Company – AFCS & Advance Research Engineer (July 2023 - Preset)

Autothrottle (Entire flight coverage, including landing) - Certified

- Analyze linear analysis results to check stability in the flight envelop and gains development.
- Aid in autothrottle closed loop non-linear simulation development, execution and analysis to better system performance, and assess aircraft stability in the flight envelope.
- Execute integrated test procedure in test facility to assess software performance and hardware in the loop integration.
- Converse with the pilots and analyze flight test HSDB and IADS telemetry data to troubleshoot, improve and develop the system.
- Perform AFCS, Nosewheel Steering, and autothrottle regression testing for aircraft safety of flight clearence.

Emergency Autoland (EAL) (Co-Lead) – In Certification, deliverables submitted to FAA

- Review and help revise design requirements document for Garmin software development.
- Create and execute integrated laboratory test plans for software integration and development.
- Create, execute and integrate EALCU (Emergency Autoland Control Unit - flaps, landing and autobrake deployement)
- Develop and execute closed loop non-linear simulations for EAL landing performance analysis and gains tuning. Includes centerline tracking, flare, crab maneuver, derotation, rollout and GPS centerline tracking with autobrake application.
- Create and execute certification laboratory test plan (witnessed by FAA and DET) - Certification deliverable
- Develop and execute EAL safety of flight test procedure to check flap, landing gear deployment, and autobraking.
- Support flight test operations by attending debriefs, pilot software familiarization, and addressing concerns in subsequent software releases.

HondaJet Echelon (Adhering to Part 25 guidelines) – In development

- Create closed loop non-linear simulation for AFCS development, including inner and outerloop gains development.
- Create stall warning and protection system preliminary requirements document and laboratory test plan. Adhering to ARP4754 guidelines in conjunction with DO-178C and ARP4761.

CREATe Cavitation Lab – Graduate Research Assistant (December 2021 – May 2023)

- Develop a left ventricle mock circulation loop to simulate and test cardiac pumps.
- Use data acquisition techniques to obtain pressure, flow data, and PIV (Particle Image Velocimetry).
- Employ data analysis techniques to study flow structure inside the left ventricle.

Graduate Teaching Assistant (Virginia Tech) (Jan 2022 – May 2023)

Create, grade assignments/projects, and host office hours.

- General Engineering – Teach MATLAB, SolidWorks, ethics, and engineering design to students.
- Propellers and Turbines – Teach Propeller selection and design, lifting line theory, powerplant matching, blade momentum theory, vortex lattice methods, propeller performance curves, actuator disk theory, and cavitation.

Advanced Propulsion and Power Lab (Research Volunteer) (May 2020-May 2021)

- Study cutting-edge multi-property measurements of vector velocity, temperature, and density in varying flow environments using Filtered Rayleigh scattering (FRS).
- Design, build, and test an acoustic enclosure for high-power steady-state lasers, prone to acoustic interference in the presence of high amplitude and low-frequency noise sources.

## COURSEWORK

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| Data Analysis in Fluid Dynamics | Boundary Layer Theory | Adv Aero/Hydrodynamics | Applied Machine Learning |
| Data Structures & Algorithms | Fluid Flows in Nature | Statics for Engineers | Booster Design |
| Applied CFD | Propellers & Turbines | Vehicle Dynamics & Control | Vehicle Propulsion |